

NIEL ABHISHEK J DAVID

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PROFESSIONAL SUMMARY

Software Engineer with experience developing AI-powered backend systems, REST APIs, cloud-native applications, and scalable data solutions across IISc Bangalore and Volvo Group. Skilled in Java, Python, Spring Boot, FastAPI, SQL, Docker, AWS, and distributed system design with a strong interest in large-scale backend engineering and AI-driven applications.

EXPERIENCE

- ❖ **Software Development Intern | Indian Institute of Science** *Jan 2025 – Jun 2025*
 - Developed backend services using **FastAPI, Firebase, WebSockets, and REST APIs** for an AI-powered Smart Mirror deployed on NVIDIA Jetson Nano and designed the **Digital Wardrobe** module with Firebase Authentication, Firestore, and Cloud Storage for secure profile and wardrobe management.
 - Integrated **React, OpenCV, Face Recognition, and Gemini LLM** for real-time computer vision, voice interaction, outfit analysis, and personalized recommendations.
- ❖ **Java Full Stack Developer | Jspiders Training & Development Centre** *Sept 2025 – Feb 2026*
 - Developed backend applications using **Java, Servlets, Spring IoC, Spring MVC, Spring Boot, Spring Security, and Spring Microservices** through hands-on development.
 - Built **REST APIs, CRUD operations, authentication/authorization, database integration, and layered applications** using Java and relational databases and applied **Git, GitHub, Maven, Docker, AWS, and CI/CD practices** to develop, version, containerize, and deploy Java applications.
- ❖ **Data Analyst | Volvo Group India Pvt Ltd** *March 2026 – Present*
 - Developed Python (Pandas/NumPy) and SQL data pipelines on Azure Databricks to automate data cleansing and preprocessing for aftermarket planning systems and designed automated ETL workflows using Power Automate, Dataverse, and Azure services to streamline cross-system enterprise data integration.
 - Built Power BI dashboards and automated reports integrated with SQL, Dataverse, and SharePoint for KPI reporting and operational insight.

PROJECTS

- ❖ **FitTrack Pro**
Tech Stack: Java, Spring Boot, Spring Security, Microservices, JWT, Hibernate, MySQL, React.js, Docker, MapStruct
 - Architected and developed a production-ready fitness management platform featuring secure authentication, role-based authorization, workout scheduling, exercise management, progress analytics, and personalized dashboards using Spring Boot and React and normalized MySQL schemas, optimized SQL queries and indexing strategies, and developed scalable REST APIs using Spring Boot and Hibernate.
 - Designed scalable RESTful APIs with layered architecture, Hibernate/JPA, DTO mapping, validation, global exception handling, and optimized database relationships following SOLID design principles.
 - Integrated AI-powered workout generation, official Spotify Web API connectivity, responsive analytics dashboards, Dockerized deployment, Swagger API documentation, and automated testing to deliver a modern SaaS-style fitness application.
- ❖ **Digital Wardrobe**
Tech Stack: FastAPI, React.js, WebSockets, Gemini LLM, OpenCV, Face-Recognition, REST API Development
 - Architected AI-driven microservices using FastAPI and WebSockets to deliver low-latency communication between smart mirror services.
 - Implemented secure facial recognition, Firebase Authentication, and cloud-based wardrobe management using Firestore and Firebase Storage.
 - Containerized backend services with Docker and integrated Gemini LLM for personalized outfit recommendations and voice-assisted interactions.
 - Designed scalable backend services using modular architecture and REST APIs to support real-time communication across distributed components.

❖ **Fashion Recommendation System**

Tech Stack: TensorFlow, EfficientNetB3, Flask, SQLAlchemy, MLflow, React

- Developed an EfficientNetB3-based deep learning model achieving 92% accuracy for personalized fashion recommendations.
- Designed PostgreSQL schemas with SQLAlchemy for efficient storage of user profiles, measurements, and recommendation history.
- Containerized ML inference services using Docker and automated model validation and deployment workflows with GitHub Actions.

❖ **Cloud-Based Attendance System**

Tech Stack: React.js, Node.js, Firebase Auth/Cloud firestore, CI/CD, GitHub Actions

- Built a responsive school attendance system with secure login, real-time updates via Firebase Firestore, and student profile management.
- Implemented a modular backend with RBAC authentication and analytics dashboard to visualize attendance patterns.
- Cloud-native solution automating attendance tracking with secure access and real-time updates.

❖ **Network Security Firewall Solution**

Tech Stack: Python, Tornado, Flask, Distributed Caching, Intrusion Detection Systems

- Developed a rule-based intrusion detection system using Python, Flask, and Tornado to support concurrent traffic inspection and centralized security rule management.
- Built RESTful management APIs allowing administrators to dynamically configure security policies and monitor suspicious network activity.
- Optimized session management using Redis caching and containerized the entire application with Docker for portable, high-throughput deployment.

EDUCATION

❖ **B.Tech in Computer Science Engineering**

PES University, RR Campus, Bengaluru

Sept 2022 – Aug 2025

CGPA – 7.73

TECHNICAL SKILLS

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- **Programming Languages:** Python, Java, C#, JavaScript, Bash
 - **Frontend Development:** HTML, CSS, React.js
 - **Backend Development:** Servlets, Spring IoC & MVC, Spring Boot, Spring Security, Hibernate, JPA, FastAPI, Django, REST APIs, JWT Authentication, ASP.NET Core
 - **Database:** MySQL, MongoDB, Oracle SQL(PL-SQL)
 - **Machine Learning Libraries:** Pandas, NumPy, TensorFlow, Scikit-Learn
 - **Testing & API Tools:** JUnit 5, Mockito, Swagger/OpenAPI, Postman
 - **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Docker Compose, GitHub Actions, CI/CD
 - **Developer Tools:** Git, GitHub, Maven, Linux, IntelliJ IDEA, Eclipse, VS Code

CERTIFICATIONS

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- Beginner's guide to Linux Kernel Development. ([Certificate](#)) _
 - AWS educate Introduction to Cloud. ([Storage](#), [Compute](#), [Serverless](#), [Networking](#))
 - Jira Work Management Fundamentals Badge ([Atlassian University](#))